

EXPERIMENT 6

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QUESTION A:

A fruit distribution company records the number of apples and oranges sold each day. For every date, exactly one record exists for apples and one record exists for oranges. Generate a daily sales comparison report showing the difference between the number of apples sold and the number of oranges sold on the same day. The difference should be calculated as:

Difference = Apples Sold - Oranges Sold

Display the sale date and the calculated difference. The result should be sorted by sale date in ascending order.

QUERY:

```
126
127  SELECT
128      s.sale_date,
129      (SELECT sold_num FROM Sales AS s1 WHERE fruit = 'apples' AND s1.sale_date = s.sale_date) -
130      (SELECT sold_num FROM Sales AS s2 WHERE fruit = 'oranges' AND s2.sale_date = s.sale_date) AS diff
131  FROM Sales AS s
132  GROUP BY s.sale_date;
133
134  SELECT
135      sale_date,
136      SUM(CASE WHEN fruit = 'apples' THEN sold_num ELSE 0 END) -
137      SUM(CASE WHEN fruit = 'oranges' THEN sold_num ELSE 0 END) AS diff
138  FROM Sales
139  GROUP BY sale_date;
140
141
142
```

RESULT:

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	sale_date	diff
1	2020-05-04	-1
2	2020-05-02	0
3	2020-05-03	20
4	2020-05-01	2

QUESTION B:

A company stores hierarchical employee reporting information in a table. Each record represents a node in a hierarchy where:

- id represents the current node.
- p_id represents the parent node.

Classify every node into one of the following categories:

- Root: A node that does not have any parent.
- Leaf: A node that has a parent but does not have any children.
- Inner: A node that has both a parent and at least one child.

Display the node ID along with its category. Return the result sorted by node ID.

QUERY:

```

157 SELECT
158     id,
159     CASE
160         WHEN p_id IS NULL THEN 'Root'
161         WHEN id NOT IN (SELECT p_id FROM Tree WHERE p_id IS NOT NULL) THEN 'Leaf'
162         ELSE 'Inner'
163     END AS type
164 FROM Tree;
165
166
167
  
```

RESULT:



Data Output		Messages	Notifications
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		id [PK] integer	type text
1		1	Root
2		2	Inner
3		3	Leaf
4		4	Leaf
5		5	Leaf
Total rows: 5		Query complete 00:00:00.140	
		CRLF Ln 166, Col 1	